Holcombe Department of Electrical and Computer Engineering
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Lab Report
Experiment 5 Mesh and Nodal Analysis

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1) Abstract

In this fifth experiment, a circuit is provided. This circuit is first analyzed using mesh analysis. With the results collected, the current across a resistor is found. Next, the same circuit is analyzed using nodal analysis. The current over the same resistor is found. With the results of both methods on the same circuit, the results are compared to seeing if both methods, when properly applied, return the same values.

2) Introduction

The main goal of the experiment is to showcase two methods of circuit analysis. These methods, mesh analysis, and nodal analysis, are key tools for solving circuits. The ability to apply both methods to a given circuit and solve it properly will allow more complex circuits to be solved by hand. The importance of using both methods is that they should yield the same results. Once proficiency is achieved with these methods, applying the right one could allow a circuit to be solved without the use of simulation software.

3) Theory and Procedure

Circuit Analysis requires understanding and application of the fundamental laws of circuits. Mesh and Nodal analysis builds on these fundamental laws, the first being Ohm's Law (1).

$$V = IR \tag{1}$$

Ohm's Law states that the voltage at a point in the circuit is equal to the current multiplied by the resistance. The other two laws are both Kirchhoff's Laws. The Kirchhoff Current Law (2) states that the sum of currents entering and leaving a node equal each other. The Current Law is used during nodal analysis. A 'loop' is identified, and isolated from the rest of the circuit. Then, the current law (2) is applied, and Ohm's law (1) is substituted in where appropriate.

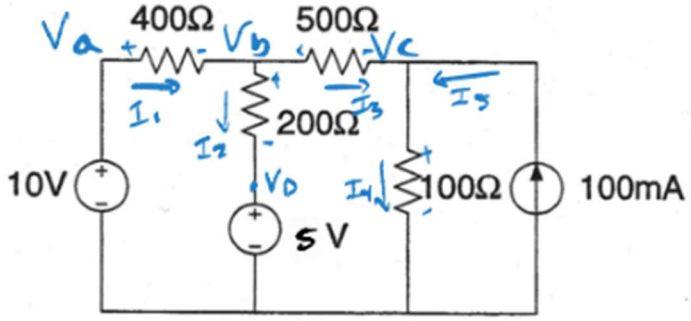
$$\Sigma I_{in} = \Sigma I_{out} \tag{2}$$

The final fundamental law, and Kirchhoff's second law, is Kirchhoff's Voltage Law (3). This law states that the sum of voltage rises and drops around a circuit equal zero. It is used during mesh analysis to create equations for the voltage drop over resistors.

$$\Sigma V_{Loop} = 0 \tag{3}$$

With these three equations, and the mesh and nodal methods of applying these equations, the given circuit can be solved with both nodal and mesh analysis.

Nodal Analysis



KVL at Node b

① $\frac{10-V_B}{400\Omega} = \frac{V_B-5}{200\Omega} + \frac{V_B-V_C}{500\Omega}$

KVL at Node C

$\frac{V_B-V_C}{500\Omega} + 0.1A = \frac{V_C}{100\Omega}$

② $V_C = \frac{V_B-V_C}{5} + 10V$

plugged into calculator ①, ②

$V_B = 7.27$

$V_C = 9.55$

$I_1 = \frac{10V-7.27V}{400\Omega} = 0.006825A$

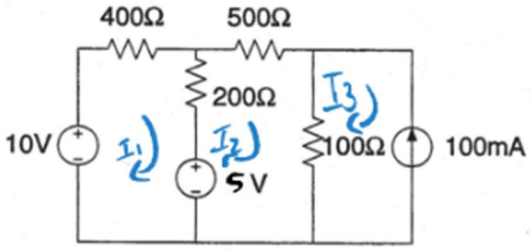
$I_2 = \frac{7.27V-5V}{200\Omega} = 0.01135\ A$

$I_3 = \frac{7.27-9.55}{500} = -0.00456A$

$I_4 = \frac{9.55V}{100} = 0.0955A$

Figure 1: Nodal Analysis Hand Calculations

Mesh Analysis



KCL loop I_1

① $10V - 5V - 400I_1 - 200(I_1 - I_3) = 0$

KCL Loop I_2

② $5V + 200(I_1 - I_3) - 500I_3 - 100(I_3 + 0.1) = 0$

① $5V - 400I_1 - 200I_1 + 200I_2 = 0$

$5V - 600I_1 = -200I_3$

$I_1 = \frac{1}{120} + \frac{1}{3}I_3$

② $5V + 200I_1 - 200I_3 - 500I_3 - 100I_3 - 10 = 0$

$-5V + 200I_1 - 800I_3 = 0$

$-5V + 200(\frac{1}{120} + \frac{1}{3}I_3) - 800I_3 = 0$

$-5V + \frac{5}{3} + \frac{200}{3}I_3 - 800I_3 = 0$

$-3.33 - 733.33I_3 = 0$

$I_3 = 0.0045A$

$I_1 = \frac{1}{120} + \frac{1}{3}(0.0045)$

$I_1 = 0.00685A$

$I_4 = I_2 - I_3 = -0.0045A + 0.1A = 0.0955A$

$I_2 = I_1 - I_3 = 0.00685A + 0.0045A = 0.01135A$

Figure 2: Mesh Analysis Hand Calculations

I. An understanding of nodal and mesh analysis is obtained.
II. The circuit is given to the students.
III. The circuit is annotated with the nodes labeled and mesh identified.
IV. Nodal Method is applied to the at point b and c, due to this including the necessary variables without adding complexity with a third equation
V. The system of equations is solved using a calculator, yielding results for Nodal Method.
VI. Mesh Analysis is applied to the circuit loops creating a system of equations.
VII. The system of equations is solved, yielding Mesh Method results.
VIII. These results are used with equation (2) to solve the remaining currents
IX. The results are compared to ensure both methods were applied correctly.

4) Discussion of Results

Applying Nodal analysis to the circuit, the two points were chosen. Point b was chosen due to its proximity to the unknown, as well as having multiple known values around it. Point c was chosen due to knowing the value of the current on the right side of the circuit. This made applying it and solving the subsequent system of equations easier. With the results for voltage solved for, Equation (1) was applied to solve for the currents. The results of these can be seen in Figure (1) in the hand calculations.

Applying Mesh analysis, loops one and two were chosen. This was due to the current in loop two which could be re-written in terms of the current in loops one and three. This simplified the calculations. After the initial calculations, the current two and current four needed to be found. As stated earlier, the current in loop two was re-written in terms of current one and three. After this was solved, current four was found with the difference between loops three and two. This second step used the KVL Equation (9) to set up the initial equations.

The results of the experiment are identical. This implies that the results for both calculations are correct due to them matching. It is significant that the results match each other, because it shows that either solution could be applied for the results. This allows choice of method, because each method has its strengths and weaknesses. An engineer could either choose the nodal or mesh method to solve a circuit. Whichever one that is chosen could be checked by the other to ensure accuracy.

5) Conclusions

This experiment is crucial in terms of circuit analysis. In the last report, KCL equation (2), and KVL equation (3) were used to solve circuits. In the second report, they were applied. Through methods such as Nodal and Mesh, more complex circuits suddenly become solvable. Each method has benefits over the other, but both a valuable tools to add to an engineer's toolbox.

References

[1] A. L. Duke and D. McAuliff, *Electrical Engineering Laboratory I*.

[2] IEEE, “IEEE SA - The IEEE Standards Association - Home,” *Ieee.org*, 2019. https://standards.ieee.org/

Appendix A: Technical Standards

IEEE/ANSI 315A-1986, NFPA 70e